

Answers

VARC

1.D	2.D	3.C	4.A	5.C	6.C	7.A	8.C
9.B	10.D	11.B	12.C	13.D	14.D	15.D	16.D
17.B	18.D	19.2	20.2	21.2431	22.3142	23.B	24.B

LRDI

25.14	26.C	27.33	28.D	29.B	30.B	31.2	32.D
33.2	34.D	35.120	36.40	37.D	38.D	39.A	40.60
41.A	42.A	43.40	44.C				

Quant

45.D	46.D	47.A	48.D	49.5	50.468	51.9	52.C
53.D	54.C	55.A	56.C	57.D	58.42	59.340	60.36
61.C	62.A	63.54	64.D	65.A	66.3		

Explanations

VARC

1. D

The primary purpose of patrimony laws is stated in the passage as being "aimed at protecting cultural property," implying that the intention behind these laws is to preserve and safeguard a country's cultural heritage. However, the paradox lies in the unintended consequence of these laws, as highlighted in the passage. The author argues that, despite the good intentions of protecting cultural property, the strict implementation of patrimony laws has led to a reduction in new archaeological discoveries. This reduction is attributed to diminished incentives for foreign entities, such as governments, NGOs, and educational institutions, to invest in overseas archaeological exploration. In other words, the very laws designed to protect cultural property end up hindering the process of making new archaeological discoveries. This underscores the tension between preserving cultural heritage and the potential negative impact on the exploration and understanding of that heritage. Option D aptly captures this point.

[VIDEO SOLUTION](#)

2. D

The author suggests that archaeological sites are important to some source countries because they can reap benefits from new archaeological discoveries, and one of the mentioned benefits is that such discoveries typically increase tourism. The passage emphasizes the economic and cultural advantages associated with tourism, which includes enhancing cultural pride and potentially attracting visitors to explore archaeological sites. Option D correctly presents this point. None of the other choices can be considered as valid inferences.

VIDEO SOLUTION

3. C

The central idea of the passage is that strict cultural property laws, although popular, may reduce incentives for foreign entities to invest in overseas archaeological exploration. The passage suggests that this reduction in incentives could be detrimental to archaeological discoveries and, consequently, to the tourism and cultural pride of source countries.

Among the given options, the only statement that would undermine the central idea is presented in Option C - it introduces the idea that there is external financial support for archaeological research in these countries. If true, then the lack of discoveries could potentially be attributed to a completely different factor/variable that the author might have failed to account for.

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4. A

Let us examine the given choices -

Option A: The passage emphasises the benefits of international collaboration and suggests that source countries could reap the benefits of new archaeological discoveries through such collaboration. Funding institutes in other countries to undertake exploration may not align with this perspective, as it involves outsourcing the exploration rather than fostering collaboration within the source country. Thus, Option A is an unlikely recommendation.

Option B: This aligns with the idea of international collaboration and suggests that the author might advise source countries to permit foreign analysis and exhibition of archaeological finds. The passage does suggest that strict cultural property laws may hinder opportunities to hold, display, and study uncovered artefacts.

Option C: We are told that strict cultural property laws might reduce incentives for foreign governments, NGOs, and educational institutions to invest in overseas exploration. Therefore, the author would support a proactive approach to encourage these entities to invest in expeditions in source countries.

Option D: The passage highlights China's shift in strategy from strict cultural property laws to embracing collaborative international archaeological research. The author suggests that this approach has led to an increase in archaeological discoveries and recognition. Hence, this would be a relevant recommendation.

Hence, Option A is the correct choice.

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5. C

Based on the discussion, the option that is NOT associated with Pinker's view of rational thinking (as well as that of the ancient Greek philosophers) is Option C - the passage suggests that while sequential reasoning is valuable, many profound human achievements come from moments of epiphany or insight rather than solely from conscious, sequential reasoning.

In relation to this thought, we are told that the emphasis on rational thought involves an understanding of the gaps in one's own knowledge [Option A] and also 'arriving at independent conclusions' [Option D]: *"Even Plato's Socrates — who anticipated many of Pinker's points by nearly 2,500 years, showing the virtue of knowing what you do not know and examining all premises in arguments, not simply trusting speakers' authority or charisma..."*

Towards the end of the passage, we are informed of an ethical and moral dimension [Option B] to rationality, which the author asserts that Pinker considers but does not elaborate on.

Hence, Option C is the correct choice.

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6. C

The passage emphasises Pinker's focus on sequential reasoning and the tools of rationality, suggesting that greater mastery of these tools can improve decision-making in various practical contexts where individuals must act on 'uncertain and shifting information.' The author's endorsement or support for Pinker's work is centred on the idea that logical reasoning "equips people with the ability to tackle challenging practical problems" [Option C].

Option A is incorrect - while the author acknowledges that rationality is seen by Pinker as a moral virtue, he adds that this role of moral and ethical education is underexplored in Pinker's work. Option B presents a very specific use case of Pinker's views and fails to capture the broader message. Option D is similarly limited in scope - the emphasis is more on the broader applicability of rationality in decision-making.

Hence, Option C is the correct choice.

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7. A

In the case of Kekulé, the discovery of the benzene structure reportedly came to him in a dream, showcasing how creative insights can emerge unexpectedly and unconsciously. Similarly, Mozart's symphonies, considered masterpieces of classical music, are often seen as products of his musical genius and creative intuition. Therefore, the examples support the notion that groundbreaking achievements in both scientific and artistic domains may involve moments of inspiration, intuition, or epiphany, challenging the idea that all significant accomplishments are the result of conscious and sequential reasoning. This aligns with the broader point that while conscious reasoning is valuable, there are also subconscious and intuitive processes at play in the generation of innovative ideas and creations. Option A correctly captures this idea.

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8. C

In the passage, the author notes that Pinker recognises rationality as both "a cognitive and moral virtue." However, the author points out that this "profoundly important" connection between rationality and morality is not thoroughly developed in Pinker's book. By bringing up the ancient Greek philosophers who, according to the text, subtly explored the connection between moral character and rationality, the author is implying that Pinker's work could benefit from a more in-depth consideration of the ethical dimension of rational behaviour. Option C accurately reflects this point - none of the other choices correctly capture the intention behind mentioning the Greek philosophers.

9. B

"The main difficulty in studying the romantics, according to him, is the lack of any "single real entity, or type of entity" that the concept "romanticism" designates."

Option B is the correct answer because it accurately captures the main difficulty highlighted in the passage when studying romanticism. The passage, particularly referencing Arthur Lovejoy, emphasizes the challenge posed by the lack of clear conceptual contours or a single, cohesive entity associated with the term "romanticism." Lovejoy's assertion that romanticism is the "scandal of literary history and criticism" underscores the difficulty in defining the boundaries and characteristics of this literary and artistic movement.

The elusive and suggestive nature of romantic aesthetics (Option A) is mentioned in the passage as a challenge, but it is not identified as the main difficulty. Similarly, the controversial and scandalous history of romantic literature (Option C) is not specified as the primary obstacle. The absence of written accounts by romantic poets and artists (Option D) is acknowledged as a challenge, but the primary difficulty highlighted in the passage is the lack of clear conceptual contours associated with romanticism.

10. D

"The most characteristic romantic commitment is to the idea that the character of art and beauty and of our engagement with them should shape all aspects of human life. Being fundamental to human existence, beauty and art should be a central ingredient not only in a philosophical or artistic life, but also in the lives of ordinary men and women."

According to the passage, the romantics rejected the idea of confining aesthetics to a specific domain separate from practical and theoretical aspects of life. Instead, they believed that aesthetics, encompassing the character of art and beauty, should permeate all aspects of human existence, not only in philosophical or artistic lives but also in the lives of ordinary men and women. Therefore Option D is the correct answer.

CAT Percentile Predictor



11. B

The passage suggests that recent studies on romanticism do not seek to overlook the differences between national romanticisms but rather attempt to characterize romanticism in terms of particular philosophical questions and concerns. The focus is on understanding romanticism without ignoring the diversity among different national expressions of it. Therefore Option B is the correct answer as it is not supported by the passage.

Option A: Although the passage acknowledges the challenges in characterizing romantic aesthetics, it also argues that it is not impossible or undesirable. Scholars recognize the difficulty but still emphasize the importance of discovering the nature of romanticism.

Option C: The passage mentions that the views of romantics on art and beauty are often found in fragments, aphorisms, and poems rather than in fully developed theoretical accounts. This emphasizes the elusive and suggestive nature of their expressions.

Option D: The passage notes that many romantics rejected the identification of aesthetics with a circumscribed domain of human life that is separated from the practical and theoretical domains of life. Instead, they believed that the character of art and beauty should shape all aspects of human life.

12. C

Option C is the correct answer because it accurately reflects the passage's explanation of why recent studies on romanticism avoid seeking "a single definition, a specific time, or a specific place." According to the passage, these studies opt to characterize romanticism in terms of "particular philosophical questions and concerns" rather than attempting to provide a singular, all-encompassing definition. The reason for this approach is to delve into the fundamental concerns of the romantics, recognizing that romanticism is a complex and multifaceted movement with diverse expressions in different nations and contexts.

The passage suggests that romanticism is not easily confined to a single, universally applicable definition due to the variety of romanticisms in different nations. Instead of discrediting or refuting Lovejoy's skepticism (Option D), recent studies acknowledge the challenges and complexities of defining romanticism but seek to understand it by focusing on the core philosophical questions and concerns that were central to the romantics' worldview.

13. D

"Those observing global climate negotiations know about the Latin American way of looking at Earth as Pachamama (Earth Mother). **They also know how such a framing is just provided lip service and is ignored in the substantive portions of the negotiations.** In *The Nutmeg's Curse*, Ghosh explains why. He shows **the extent of the vested interest in the oil economy - not only for oil-exporting countries, but also for a superpower like the US that controls oil drilling, oil prices and oil movement around the world.** Many of us know **power utilities are sabotaging decentralised solar power generation today because it hits their revenues and control.**"

From the highlighted part we can clearly infer Options A, B and C. The passage does not suggest that the decentralised characteristic of renewable energy resources like solar power is a reason for the failure of climate change policies. Instead, it mentions that power utilities may be sabotaging decentralized solar power generation because it affects their revenues and control, but it does not frame the decentralised nature of renewable energy as a cause for failure. Therefore Option D is not a reason for the failure of policies seeking to address climate change.

14. D

If non-European societies perceive the Earth as a non-living source of all resources, it contradicts the personification implied by the use of "who" for Gaia.

In the context of the passage, the author uses the word "curse" in the title, and the pronoun "who" is used for Gaia, suggesting a perspective that sees Earth as a living, conscious entity. If non-European societies do not share this perspective and view the Earth as a non-living source of resources, it challenges the appropriateness of using the pronoun "who" for Gaia in the context of their beliefs. This would make the reviewer's choice of the pronoun "who" inappropriate, given the differing perspectives on the nature of the Earth. Therefore Option D is the correct answer.

VIDEO SOLUTION

15.D

Option D cannot be directly drawn from the passage. The passage discusses the historical perspective on climate change presented in "The Nutmeg's Curse" and emphasizes the impact of colonialism on the contemporary dominant perception of nature and the environment. It suggests that there are alternative views from non-European and/or pre-colonial societies that can provide insights for environmental preservation policy makers. However, the passage does not explicitly state that academic discourses have always served the function of raising awareness about environmental preservation.

The passage connects the history of climate change with colonialism (Option A), highlights that colonial processes shaped the contemporary perception of nature and the environment (Option B) and suggests that non-European and/or pre-colonial societies hold valuable insights for environmental preservation policy makers (Option C).

VIDEO SOLUTION



16.D

"These are the two points to which Ghosh returns through examples from around the world. One, how European colonialists decimated not only indigenous populations but also indigenous understanding of the relationship between humans and Earth. Two, how this was an invasion not only of humans but of the Earth itself, and how this continues to the present day by looking at nature as a 'resource' to exploit"

The passage discusses how the colonization of the Banda islands, as presented in "The Nutmeg's Curse," is used to illustrate the broader idea that colonialism played a significant role in shaping the mindset and practices that have led to climate change. The exploitation of both indigenous populations and the Earth's resources during colonialism is portrayed as a key factor in perpetuating the mindset that views nature as a resource to be exploited, contributing to the environmental challenges faced today. Therefore, the primary purpose of discussing the colonization of the Banda islands is to highlight how colonialism represented and perpetuated the mindset that has led to climate change. Therefore Option D is the correct answer.

VIDEO SOLUTION

17.B

The Sentence best fits in Blank 3 because it directly relates to the theme introduced in before Blank 3. Before Blank 3 the passage discusses the linkages between illicit arms, organized crime, and armed conflict, and the provided sentence explains how armed conflict creates an environment favorable for organized crime to thrive. Therefore, Blank 3 is the appropriate placement for the sentence, enhancing the coherence and thematic continuity of the paragraph.

Blank 1 and 4 can be easily eliminated as the Sentence in question does not serve as a proper introductory or a concluding statement.

The provided sentence introduces a different aspect, focusing on the broader impact of armed conflict on organized crime rather than the financial dynamics discussed after Blank 2.

VIDEO SOLUTION

18.D

The sentence is most fitting in Blank 3 because it follows the idea presented in the previous sentences. Before Blank 3 the passage discusses the various ways to understand the relation of support between premises and conclusions, presenting different perspectives on how arguments can be analysed or interpreted. The sentence in question adds another layer to this discussion by suggesting that, for theoretical purposes, arguments can be considered independently of their real-world contexts. This placement enhances the overall flow of the paragraph by introducing a theoretical perspective on the nature of arguments.

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19.2

Sentence 2 doesn't fit well with the others, which focus on the language Bo and its last native speaker, Boa Senior. The other sentences provide information about the language, its history, and the last fluent speaker, creating a coherent narrative. Sentence 2 introduces a different topic about the decline of the indigenous population without directly contributing to the discussion about the tribal language and its last speaker.

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20.2

Sentence 2 is the odd one out because it does not directly contribute to the theme of a shift in hiring priorities towards soft skills. Sentences 1, 3, 4, and 5 all discuss changes in hiring practices, the importance of soft skills, and the response to the pandemic, creating a coherent narrative. In contrast, Sentence 2 introduces a more general idea about redefining the ideal candidate without specifically addressing the shift towards prioritizing soft skills, making it the odd one out in the context of the paragraph's focus.

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21.2431

The correct order is 2-4-3-1

Sentence 2 introduces the immediate impact of the printing press on how people learned in the scribal culture, emphasizing the shift from listening and memorization to a new learning process.

Sentence 4 builds on this by explaining the transformation brought about by the printing press, highlighting the shift from listening and memorization to reading and the newfound ability to learn privately.

Sentence 3 provides additional context, emphasizing that the transformation from listeners to readers was a complex social and cultural phenomenon that became more widespread during the industrial era.

Sentence 1 concludes the paragraph by noting that despite technological advancements such as film, radio, and television, centuries later, formal learning is still predominantly based on reading. This brings the historical context to the present, concluding the narrative cohesively.

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22. 3142

Sentence 3 introduces the issue, highlighting that e-waste contains valuable materials that are not being efficiently recycled. This sets context for the discussion to be followed.

Now if we consider Sentences 1, 2 and 4, we can see that Sentence 2 and 4 are referencing a "She" and "Her". Considering these 3 sentences, they can only reference Veena Sahajwalla. Therefore Sentence 1 must follow Sentence 3.

Now Sentence 1 states that she has a new way of solving the problem. Sentence 4 provides details about Sahajwalla's plan, stating that she intends to build microfactories to extract valuable materials from e-waste. Therefore Sentence 4 must be following Sentence 1. Sentence 2 logically follows Sentence 4 by elaborating on Sahajwalla's vision, explaining how automated drones and robots will be utilized in the process of extracting metals from e-waste, providing a clear and detailed picture of the proposed solution.

Therefore the correct order is 3-1-4-2.

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23. B

The passage contrasts the 1950s, where perfectionism meant conforming to societal norms, with contemporary times, where individuals feel pressured to stand out and gain attention through unique style and wit. This evolution from conformism to non-conformism is a key point in the passage, making Option B the most accurate summary. It effectively captures the changing nature of perfectionism in response to multifarious and contradictory societal expectations over the decades.

Option A focuses on the media's role and people adhering to ideals, which is not the primary emphasis of the passage that highlights the evolution of societal expectations over time.

Option C suggests tension and conflict related to the changing idea of perfection, but the passage emphasises the historical shift in expectations rather than conflict.

Option D overgeneralizes by stating that people are willing to go to any lengths to attract attention, which is not explicitly supported by the passage that highlights the changing nature of perfectionism.

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24. B

The passage describes how life on the island is gradually improving, particularly for birds like Antarctic prions and white-headed petrels. It highlights that these birds are increasing in numbers as pests are controlled on the island. The absence of pests allows the birds to return, breed, and contribute to the ecosystem positively. The passage also mentions how bird droppings add nutrients to the soil, supporting plant growth. Overall, Option B effectively conveys the central theme of the passage - the revival and improvement of life on the island due to the absence of pests and the positive impact on birds and plants.

Option A incorrectly suggests an increase in the number of predatory birds, which contradicts the positive developments mentioned in the passage.

Option C is incorrect as the passage doesn't explicitly state that this protection is the primary cause of the positive changes.

Option D is more general and doesn't specifically address the absence of pests as a crucial factor in the positive transformation mentioned in the passage.

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LRDI

Explanation [25 - 29]:

Let us assume, A is the total number of AC's sold

=> From the information that the total number of ACs sold in the city, 25% were of Window variant => Window AC's = $A/4$ and Split AC's = $3A/4$

Now, let us assume B is the total number of inverter ACs

=> From the information that among the Inverter ACs sold, 20% were of Window variant. => Window Inverter AC's = $B/5$ and Window Non-Inverter AC's = $4B/5$

Total (A)			
Split ($3A/4$)		Window ($A/4$)	
Inv ($4B/5$)	Non-Inv	Inv ($B/5$)	Non-Inv

From - Condition-3

=> $A/4 - B/5 = 6$ and $4B/5 = 36$ => $B = 46$ and $A = 60$.

Total = 60			
Split = 45		Window = 15	
Inv = 36	Non-Inv = 9	Inv = 9	Non-Inv = 6

Now, from condition-6

a) D1 & D4 sold "0" window Non-inverter ACs => D2 & D3 sold 6 window non-inverter ACs, it is given that D2 sold twice as many as D3 => D2 sold 4 and D3 sold 2 ACs of this type.

From condition-2

b) Let us assume, D1 sold "x" window inverter ACs => Number of split inverter ACs sold is $13-x$

From condition-4

c) Number of split ACs sold by D1 will be "2x"

From condition-5

d) Let us assume 'y' is the number of window ACs sold by D3 & D4 => D2 sold 3y ACs of this type.

From condition-7

e) Let us assume 'z' is the number of split inverter ACs sold by D3 and D4 => D2 sold 2z ACs of this type.

Let us use a, b, c, d, and e make a table:

D1 Total =			
Split =		Window = x	
Inv = 13-x	Non-Inv =	Inv = x	Non-Inv = 0

D2 Total =			
Split =		Window = 3y	
Inv = 2z	Non-Inv =	Inv =	Non-Inv = 4

D3 Total =			
Split =		Window = y	
Inv = z	Non-Inv = 3	Inv =	Non-Inv = 2

D4 Total =			
Split =		Window = y	
Inv = z	Non-Inv =	Inv =	Non-Inv = 0

We know that the total number of window ACs is 15

=> $x + 3y + y + y = 15$ => $x + 5y = 15$, also x and y should be greater than or equal to 2 from condition-1

=> x = 5 and y = 2 is the only solution.

Filling this in the table:

D1 Total =			
Split =		Window = 5	
Inv = 8	Non-Inv =	Inv = 5	Non-Inv = 0

D2 Total =			
Split =		Window = 6	
Inv = 2z	Non-Inv =	Inv = 2	Non-Inv = 4

D3 Total =			
Split =		Window = 2	
Inv = z	Non-Inv = 3	Inv = 0	Non-Inv = 2

D4 Total =			
Split =		Window = 2	
Inv = z	Non-Inv =	Inv = 2	Non-Inv = 0

Now, Number of split inverter ACs is 36

=> $8 + 2z + z + z = 36$ => $4z = 28$ => $z = 7$.

Filling this and using (5), the number of split AC's sold by D1 is $2 \times 5 = 10$.

D1 Total = 15			
Split = 10		Window = 5	
Inv = 8	Non-Inv = 2	Inv = 5	Non-Inv = 0

D2 Total =			
Split =		Window = 6	
Inv = 14	Non-Inv =	Inv = 2	Non-Inv = 4

D3 Total = 12			
Split = 10		Window = 2	
Inv = 7	Non-Inv = 3	Inv = 0	Non-Inv = 2

D4 Total =			
Split =		Window = 2	
Inv = 7	Non-Inv =	Inv = 2	Non-Inv = 0

25.14

From the table, we see that 14 split inverter ACs are sold.

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26.C

Total = 60			
Split = 45		Window = 15	
Inv = 36	Non-Inv = 9	Inv = 9	Non-Inv = 6

From this table, we see that total number of non-inverter ACs is $9 + 6 = 15$.

Required percentage is 15 out of 60 => 25%.

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27.33

Total number of ACs sold by D2 and D4 = $60 - D1 - D3 = 60 - 15 - 12 = 33$.

28. D

We see that D1 & D3 sold 27 ACs together which is less than $60 - 27 = 33$ sold by D2 & D4.

=> Option-D is definitely false.

29. B

If D3 and D4 sold equal number of AC's, the table will look as follows:

D1 Total = 15			
Split = 10		Window = 5	
Inv = 8	Non-Inv = 2	Inv = 5	Non-Inv = 0

D2 Total = 21			
Split = 15		Window = 6	
Inv = 14	Non-Inv = 1	Inv = 2	Non-Inv = 4

D3 Total = 12			
Split = 10		Window = 2	
Inv = 7	Non-Inv = 3	Inv = 0	Non-Inv = 2

D4 Total = 12			
Split = 10		Window = 2	
Inv = 7	Non-Inv = 3	Inv = 2	Non-Inv = 0

Number of non-inverter ACs sold is $1 + 4 = 5$

30. B

It is given that none of the streets has more than one team traveling along it in any direction at any point in time (point 1), which implies at 9.00 hrs, all 4 teams have chosen different roots from the starting point.

It is also known that Teams 2 and 3 are the only ones in stations E and D respectively at 10:00 hrs, and Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.

It is only possible when Team 2 traveled (A-E) via F, and Team 3 reached station D via station C.

It is also known that Teams 1 and 3 are the only ones in Station E at 10:30 hrs, and Team 4 never passes through Stations B, D, or F. Hence, Team 1 must have chosen the (A-B) root at the starting point, and Team 4 has chosen the (A-E) root at 9.00 hrs.

Hence, Team 1 will reach B at 9.30, and come to A at 10.00 hrs. After that, they will go to E at 10.30 hrs.

Since Team 4 never passes through stations B, D, or F. Team 4 only can pass through stations A, E, and C.

Hence, the roots of team 4 to reach station E at 11.30 will be (A-E-A-C-A-E) or (A-E-A-E-A-E).

Since team 1 is already traveling to E from A at 10.00 hrs, at that time team 4 can't choose the same route. Hence, the final route for team 4 to reach E at 11.30 is (A-E-A-C-A-E), and at 12.00 hrs, team 4 will come back to station A.

Hence, the complete route diagram for team 4 is (A-E-A-C-A-E-A)

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E			
2	A	F	E				
3	A	C	D				
4	A	E	A	C	A	E	A

We can see that team 1 is at station E at 10.30 hrs, and they will reach station B at 11.30 hrs, which is only possible when they travel to B via A.

Hence, the complete route diagram for team 1 is (A-B-A-E-A-B-A). It is also known that Teams 1 and 3 are the only ones in station E at 10:30 hrs.

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E				
3	A	C	D	E			
4	A	E	A	C	A	E	A

The only possible root for Team 2 at 10.00 hrs is from E to F since they can't choose E to D because Team 3 is already on this route. Since team 3 has to reach A at 12.00. The only possible combination for team 3 is E-D-C-A

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F			
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

Now the roots for team 2 going back to A is from F at 10.30 hrs (F-A-F-A) or (F-E-F-A).

Hence, the final table is given below:

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F	A/E	F	A
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

From the table, we can see that among the options station E is visited the largest number of times.

[VIDEO SOLUTION](#)



31.2

It is given that none of the streets has more than one team traveling along it in any direction at any point in time (point 1), which implies at 9.00 hrs, all 4 teams have chosen different roots from the starting point.

It is also known that Teams 2 and 3 are the only ones in stations E and D respectively at 10:00 hrs, and Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.

It is only possible when Team 2 traveled (A-E) via F, and Team 3 reached station D via station C.

It is also known that Teams 1 and 3 are the only ones in Station E at 10:30 hrs, and Team 4 never passes through Stations B, D, or F. Hence, Team 1 must have chosen the (A-B) root at the starting point, and Team 4 has chosen the (A-E) root at 9.00 hrs.

Hence, Team 1 will reach B at 9.30, and come to A at 10.00 hrs. After that, they will go to E at 10.30 hrs.

Since Team 4 never passes through stations B, D, or F. Team 4 only can pass through stations A, E, and C.

Hence, the roots of team 4 to reach station E at 11.30 will be (A-E-A-C-A-E) or (A-E-A-E-A-E).

Since team 1 is already traveling to E from A at 10.00 hrs, at that time team 4 can't choose the same route. Hence, the final route for team 4 to reach E at 11.30 is (A-E-A-C-A-E), and at 12.00 hrs, team 4 will come back to station A.

Hence, the complete route diagram for team 4 is (A-E-A-C-A-E-A)

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E			
2	A	F	E				
3	A	C	D				
4	A	E	A	C	A	E	A

We can see that team 1 is at station E at 10.30 hrs, and they will reach station B at 11.30 hrs, which is only possible when they travel to B via A.

Hence, the complete route diagram for team 1 is (A-B-A-E-A-B-A). It is also known that Teams 1 and 3 are the only ones in station E at 10:30 hrs.

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E				
3	A	C	D	E			
4	A	E	A	C	A	E	A

The only possible root for Team 2 at 10.00 hrs is from E to F since they can't choose E to D because Team 3 is already on this route. Since team 3 has to reach A at 12.00. The only possible combination for team 3 is E-D-C-A

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F			
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

Now the roots for team 2 going back to A is from F at 10.30 hrs (F-A-F-A) or (F-E-F-A).

Hence, the final table is given below:

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F	A/E	F	A
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

From the table, we can see that the teams have passed through B 2 times in this given period.

[VIDEO SOLUTION](#)

32. D

It is given that none of the streets has more than one team traveling along it in any direction at any point in time (point 1), which implies at 9.00 hrs, all 4 teams have chosen different roots from the starting point.

It is also known that Teams 2 and 3 are the only ones in stations E and D respectively at 10:00 hrs, and Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.

It is only possible when Team 2 traveled (A-E) via F, and Team 3 reached station D via station C.

It is also known that Teams 1 and 3 are the only ones in Station E at 10:30 hrs, and Team 4 never passes through Stations B, D, or F. Hence, Team 1 must have chosen the (A-B) root at the starting point, and Team 4 has chosen the (A-E) root at 9.00 hrs.

Hence, Team 1 will reach B at 9.30, and come to A at 10.00 hrs. After that, they will go to E at 10.30 hrs.

Since Team 4 never passes through stations B, D, or F. Team 4 only can pass through stations A, E, and C.

Hence, the roots of team 4 to reach station E at 11.30 will be (A-E-A-C-A-E) or (A-E-A-E-A-E).

Since team 1 is already traveling to E from A at 10.00 hrs, at that time team 4 can't choose the same route. Hence, the final route for team 4 to reach E at 11.30 is (A-E-A-C-A-E), and at 12.00 hrs, team 4 will come back to station A.

Hence, the complete route diagram for team 4 is (A-E-A-C-A-E-A)

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E			
2	A	F	E				
3	A	C	D				
4	A	E	A	C	A	E	A

We can see that team 1 is at station E at 10.30 hrs, and they will reach station B at 11.30 hrs, which is only possible when they travel to B via A.

Hence, the complete route diagram for team 1 is (A-B-A-E-A-B-A). It is also known that Teams 1 and 3 are the only ones in station E at 10:30 hrs.

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E				
3	A	C	D	E			
4	A	E	A	C	A	E	A

The only possible root for Team 2 at 10.00 hrs is from E to F since they can't choose E to D because Team 3 is already on this route. Since team 3 has to reach A at 12.00. The only possible combination for team 3 is E-D-C-A

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F			
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

Now the roots for team 2 going back to A is from F at 10.30 hrs (F-A-F-A) or (F-E-F-A).

Hence, the final table is given below:

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F	A/E	F	A
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

From the table, we can see that at 10.15 hrs, team 3 is travelling from station D to station E.

The correct option is D

[VIDEO SOLUTION](#)

33.2

It is given that none of the streets has more than one team traveling along it in any direction at any point in time (point 1), which implies at 9.00 hrs, all 4 teams have chosen different roots from the starting point.

It is also known that Teams 2 and 3 are the only ones in stations E and D respectively at 10:00 hrs, and Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.

It is only possible when Team 2 traveled (A-E) via F, and Team 3 reached station D via station C.

It is also known that Teams 1 and 3 are the only ones in Station E at 10:30 hrs, and Team 4 never passes through Stations B, D, or F. Hence, Team 1 must have chosen the (A-B) root at the starting point, and Team 4 has chosen the (A-E) root at 9.00 hrs.

Hence, Team 1 will reach B at 9.30, and come to A at 10.00 hrs. After that, they will go to E at 10.30 hrs.

Since Team 4 never passes through stations B, D, or F. Team 4 only can pass through stations A, E, and C.

Hence, the roots of team 4 to reach station E at 11.30 will be (A-E-A-C-A-E) or (A-E-A-E-A-E).

Since team 1 is already traveling to E from A at 10.00 hrs, at that time team 4 can't choose the same route. Hence, the final route for team 4 to reach E at 11.30 is (A-E-A-C-A-E), and at 12.00 hrs, team 4 will come back to station A.

Hence, the complete route diagram for team 4 is (A-E-A-C-A-E-A)

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E			
2	A	F	E				
3	A	C	D				
4	A	E	A	C	A	E	A

We can see that team 1 is at station E at 10.30 hrs, and they will reach station B at 11.30 hrs, which is only possible when they travel to B via A.

Hence, the complete route diagram for team 1 is (A-B-A-E-A-B-A). It is also known that Teams 1 and 3 are the only ones in station E at 10:30 hrs.

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E				
3	A	C	D	E			
4	A	E	A	C	A	E	A

The only possible root for Team 2 at 10.00 hrs is from E to F since they can't choose E to D because Team 3 is already on this route. Since team 3 has to reach A at 12.00. The only possible combination for team 3 is E-D-C-A

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F			
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

Now the roots for team 2 going back to A is from F at 10.30 hrs (F-A-F-A) or (F-E-F-A).

Hence, the final table is given below:

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F	A/E	F	A
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

From the table, we can see that team 4 passed station E 2 times in a day

[VIDEO SOLUTION](#)

34.D

It is given that none of the streets has more than one team traveling along it in any direction at any point in time (point 1), which implies at 9.00 hrs, all 4 teams have chosen different roots from the starting point.

It is also known that Teams 2 and 3 are the only ones in stations E and D respectively at 10:00 hrs, and Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.

It is only possible when Team 2 traveled (A-E) via F, and Team 3 reached station D via station C.

It is also known that Teams 1 and 3 are the only ones in Station E at 10:30 hrs, and Team 4 never passes through Stations B, D, or F. Hence, Team 1 must have chosen the (A-B) root at the starting point, and Team 4 has chosen the (A-E) root at 9.00 hrs.

Hence, Team 1 will reach B at 9.30, and come to A at 10.00 hrs. After that, they will go to E at 10.30 hrs.

Since Team 4 never passes through stations B, D, or F. Team 4 only can pass through stations A, E, and C.

Hence, the roots of team 4 to reach station E at 11.30 will be (A-E-A-C-A-E) or (A-E-A-E-A-E).

Since team 1 is already traveling to E from A at 10.00 hrs, at that time team 4 can't choose the same route. Hence, the final route for team 4 to reach E at 11.30 is (A-E-A-C-A-E), and at 12.00 hrs, team 4 will come back to station A.

Hence, the complete route diagram for team 4 is (A-E-A-C-A-E-A)

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E			
2	A	F	E				
3	A	C	D				
4	A	E	A	C	A	E	A

We can see that team 1 is at station E at 10.30 hrs, and they will reach station B at 11.30 hrs, which is only possible when they travel to B via A.

Hence, the complete route diagram for team 1 is (A-B-A-E-A-B-A). It is also known that Teams 1 and 3 are the only ones in station E at 10:30 hrs.

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E				
3	A	C	D	E			
4	A	E	A	C	A	E	A

The only possible root for Team 2 at 10.00 hrs is from E to F since they can't choose E to D because Team 3 is already on this route. Since team 3 has to reach A at 12.00. The only possible combination for team 3 is E-D-C-A

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F			
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

Now the roots for team 2 going back to A is from F at 10.30 hrs (F-A-F-A) or (F-E-F-A).

Hence, the final table is given below:

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F	A/E	F	A
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

From the table, we can see that 2 teams (teams 3 and 4) have passed through station C on the given day.

The correct option is D

[VIDEO SOLUTION](#)

Explanation [35 - 39]:

Given that in every month, both online and offline registration numbers were multiples of 10.

From (2), in Jan, the number of offline registrations was twice that of online registrations.

=> If x is number of online registrations => $2x$ is the number of offline registrations => $3x$ is the total number of registrations.

According to the data given in the table => $3x$ should lie between the minimum and maximum total number of registrations. => $x = 40$ (as x should also be a multiple of 10)

=> In Jan => (40,80) are the online and offline registrations respectively.

Similarly from (3) => In Apr (80,40) are the online and offline registrations respectively.

From-5, the number of online registrations is highest in may => In may there are 100 online registrations. The lowest possible number of offline registrations is 30 and maximum possible total registrations is 130 => In May (100,30) are the online and offline registrations respectively.

Let us assume, ' x ' to be the number of offline registrations in May = number of online registrations in March.

Let us capture all this data in a table:

Month	Online	Offline	Total
Jan	40	80	120
Feb	y	x	
Mar	x	z	
Apr	80	40	120
May	100	30	130

From the table given in the question, 50 is the median for Offline data

=> x should lie between 50 and 80 (included)

For 80 to be the median for the online data => y lie between 80 and 100 (included).

Now, consider Feb => Minimum value of $y + x = 80 + 50 = 130$ (which is the maximum value possible of the total possible registrations)

=> $x = 50$ and $y = 80$

Since, 110 is the minimum number of total registrations, the only possibility is in March => $50 + z = 110$ => $z = 60$.

Now, filling the complete table we get,

Month	Online	Offline	Total
Jan	40	80	120
Feb	80	50	130
Mar	50	60	110
Apr	80	40	120
May	100	30	130

35. 120

The total number of registrations in April is 120.

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36.40

The number of online registrations in Jan is 40.

VIDEO SOLUTION

37.D

1) In May, there are 30 offline registrations (smallest) => True

2) In Mar, we have smallest number of total registrations => False.

VIDEO SOLUTION

38.D

The number of offline registrations in Feb is 50.

VIDEO SOLUTION

39.A

Total registrations in Jan = Apr = 120 and Feb = May = 130.

VIDEO SOLUTION

Explanation [40 - 44]:

It is given that there are only three female students - Amala, Koli, and Rini - and only three male students - Biman, Mathew, and Shyamal - in a course.

It is also known that the aggregate score in the course is a weighted average of the two components, with the weights being positive and adding to 1.

Let the project score component be x , which implies the test score component will be $(1-x)$. The projects are done in groups of two, with each group consisting of a female and a male student, which implies there are three groups for the project. It is also known that both the group members obtain the same score in the project. The score obtained in the project is 40, 60, and 80, respectively.

Therefore, we can say that each female student will consist of a different group, and no two male students or female students will be in the same group.

For the test scores, there are six scores given for six students among which four are distinct and the remaining two are average scores, which is 60. It is also known that the maximum score possible is 80, and the minimum score is 40.

Hence, the distinct scores are 80, 70, 50, and 40 (since all the test scores are multiple of 10), and the remaining two scores are 60, and 60, respectively.

From point 3, we know that Amala's score in the project was double that of Koli in the same, but Koli scored 20 more than Amala in the test. Hence, we can say the score obtained by Amala in the project is 80, and the score obtained by Koli is 40, which implies the score obtained by Rini in the project is 60. Now, Koli scored 20 more than Amala in the test, which implies the score obtained by Koli can be either 80, 70, or 60.

The score obtained by them is given below:

Students	Test scores	Project scores
Amala	40/50/60	80
Koli	60/70/80	40
Rini		60
Biman		
Mathew		
Shyamal		

It is known that Amala had the highest aggregate score, and Shyamal scored the second highest on the test. He scored two more than Koli, but two less than Amala in the aggregate.

Hence, the score obtained by Shyamal in the test is 70, which implies Koli can't score 70 in the test => Amala can't score 50 in the test.

Students	Test scores	Project scores
Amala	40/60	80
Koli	60/80	40
Rini		60
Biman		
Mathew		
Shyamal	70	

It is given that Shyamal scored two more than Koli, but two less than Amala in the aggregate. Hence, the aggregate score of Amala is 4 more than Koli. It is also known that Amala had the highest aggregate score.

Case 1: The test score of Amala is 40

Students	Test scores	Project scores	Aggregate score
Amala	40	80	$40(1-x)+80x$
Koli	60	40	$60(1-x)+40x$
Rini		60	
Biman			
Mathew			
Shyamal	70		

Therefore, $40(1-x)+80x = 60(1-x)+40x+4$

=> $60x = 24$

$$\Rightarrow x = 0.4$$

Hence, the aggregate score obtained by Amala is $40(1-0.4)+80*4 = 56$

The minimum aggregate score of Shyamal is $70(1-0.4)+40*0.4 = 58$, which is greater than Amala.

Hence, **Case 1 is not possible.**

Hence, the table is given below:

Students	Test scores	Project scores	Aggregate score
Amala	60	80	$60(1-x)+80x$
Koli	80	40	$80(1-x)+40x$
Rini		60	
Biman			
Mathew			
Shyamal	70		

Therefore, $60(1-x)+80x = 80(1-x)+40x+4$

$$\Rightarrow 60+20x = 84-40x$$

$$\Rightarrow 60x = 24 \Rightarrow x = 0.4$$

Hence, the aggregate score of Amala is $60(1-0.4)+80*0.4 = 68$, which implies the aggregate score of Shyamal is $(68-2) = 66$

Hence, the score obtained by Shyamal in Project is $(66-70*(0.6))/0.4 = 60$.

It is also known that Biman scored second lowest in the test, which implies the score of Biman in the test is 50, and he scored the lowest in the aggregate. It is also known that Mathew scored more than Rini in the project, but less than her in the test. Hence, Mathew scored 80 in the project (since Rini scored 60 in the project), and Biman scored 40 in the project.

Similarly, Rini Scored more than Mathew on the test, which implies the score obtained by Rini is 60, and the score obtained by Mathew is 40 in the test.

Hence, the final table will look like this:

Students	Test scores (T)	Project scores(P)	Aggregate score (T*0.6+P*0.4)	Project Pair
Amala	60	80	68	Amala, Mathew
Koli	80	40	64	Koli, Biman
Rini	60	60	60	Rini, Shyamal
Biman	50	40	46	Biman, Koli
Mathew	40	80	56	Mathew, Amala
Shyamal	70	60	66	Shyamal, Rini

40. **60**

Hence, the score obtained by Rini in the project is 60

[VIDEO SOLUTION](#)



41. **A**

Hence, the weight of the test component is 0.6

The correct option is A

[VIDEO SOLUTION](#)

42. **A**

Hence, the maximum aggregate score obtained is 68. The correct option is A

[VIDEO SOLUTION](#)

43. **40**

Hence, the score obtained by Mathew in the test is 40

[VIDEO SOLUTION](#)

44. **C**

From the table, we can see that (Amala, Mathew), (Koli, Biman), and (Shyama, Rini) are the three groups for the project.

Hence, the correct option is C

[VIDEO SOLUTION](#)

Quant

45. **D**

It is given that $x^8 + \left(\frac{1}{x}\right)^8 = 47$, which can be written as:

$$\Rightarrow (x^4)^2 + \left(\frac{1}{x^4}\right)^2 = 47$$

$$\Rightarrow \left(x^4 + \frac{1}{x^4}\right)^2 - 2 \cdot x^4 \cdot \frac{1}{x^4} = 47$$

$$\Rightarrow \left(x^4 + \frac{1}{x^4}\right)^2 = 49$$

$$\Rightarrow x^4 + \frac{1}{x^4} = 7$$

Similarly, $x^4 + \frac{1}{x^4} = 7$ can be expressed as:

$$\Rightarrow \left(x^2\right)^2 + \left(\frac{1}{x^2}\right)^2 = 7$$

$$\Rightarrow \left(x^2 + \frac{1}{x^2}\right)^2 - 2 \cdot x^2 \cdot \frac{1}{x^2} = 7$$

$$\Rightarrow \left(x^2 + \frac{1}{x^2}\right)^2 = 9$$

$$\Rightarrow x^2 + \frac{1}{x^2} = 3$$

By the same logic, we get $x + \frac{1}{x} = \sqrt{5}$

$$\text{Now, } x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3 \cdot x \cdot \frac{1}{x} \left(x + \frac{1}{x}\right)$$

$$\Rightarrow x^3 + \frac{1}{x^3} = (\sqrt{5})^3 - 3\sqrt{5} = 2\sqrt{5}$$

By the same logic, we can say that

$$\Rightarrow x^9 + \frac{1}{x^9} = \left(x^3 + \frac{1}{x^3}\right)^3 - 3 \cdot x^3 \cdot \frac{1}{x^3} \left(x^3 + \frac{1}{x^3}\right)$$

$$\Rightarrow x^9 + \frac{1}{x^9} = (2\sqrt{5})^3 - 3(2\sqrt{5})$$

$$\Rightarrow x^9 + \frac{1}{x^9} = 40\sqrt{5} - 6\sqrt{5} = 34\sqrt{5}$$

The correct option is D

[VIDEO SOLUTION](#)



46. D

It is given that there are exactly 41 numbers, which can be expressed as the power of two, and exist between 8^m and 8^n , (where m, and n are positive integers, and $m < n$)

$$\text{Hence, } 2^{3m} < 41 \text{ numbers} < 2^{3n}$$

Since, m is a positive integer, the least value of m is 1. Therefore, $2^{3m} = 2^3$, hence, the 41 numbers between them are $2^4, 2^5, 2^6, \dots, 2^{44}$.

Then the lowest possible value of 8^n is 2^{45} . Hence, the smallest value of n is $2^{45} = 8^n \Rightarrow 2^{3n} = 2^{45} \Rightarrow n = 15$

Hence, the smallest value of $m+n$ is $(15+1) = 16$

The correct option is D

[VIDEO SOLUTION](#)

47. A

It is given that for some real numbers a and b, the system of equations $x + y = 4$ and $(a + 5)x + (b^2 - 15)y = 8b$ has infinitely many solutions for x and y.

Hence, we can say that

$$\Rightarrow \frac{a+5}{1} = \frac{b^2-15}{1} = \frac{8b}{4}$$

This equation can be used to find the value of a, and b.

Firstly, we will determine the value of b.

$$\Rightarrow \frac{b^2-15}{1} = \frac{8b}{4} \Rightarrow b^2 - 2b - 15 = 0$$

$$\Rightarrow (b - 5)(b + 3) = 0$$

Hence, the values of b are 5, and -3, respectively.

The value of a can be expressed in terms of b, which is $a + 5 = b^2 - 15 \Rightarrow a = b^2 - 20$

$$\text{When } b = 5, a = 5^2 - 20 = 5$$

$$\text{When } b = -3, a = 3^2 - 20 = -11$$

$$\text{The maximum value of } ab = (-3) \cdot (-11) = 33$$

The correct option is A

[VIDEO SOLUTION](#)

48. D

It is given that $\frac{1}{2}$, $\frac{\log_3(2^x-9)}{\log_3 4}$, and $\frac{\log_5(2^x+\frac{17}{2})}{\log_5 4}$ are in an arithmetic progression.

$\frac{\log_3(2^x-9)}{\log_3 4}$ can be written as $\log_4(2^x - 9)$, and $\frac{\log_5(2^x+\frac{17}{2})}{\log_5 4}$ can be written as $\log_4(2^x + \frac{17}{2})$

$$\text{Hence, } 2\log_4(2^x - 9) = \frac{1}{2} + \log_4(2^x + \frac{17}{2})$$

$\frac{1}{2}$ can be written as $\log_4 2$.

Therefore,

$$\Rightarrow 2 \log_4 (2^x - 9) = \log_4 2 + \log_4 (2^x + \frac{17}{2})$$

$$\Rightarrow \log_4 (2^x - 9)^2 = \log_4 2 (2^x + \frac{17}{2})$$

$$\Rightarrow (2^x - 9)^2 = 2 (2^x + \frac{17}{2})$$

$$\Rightarrow 2^{2x} - 18 \cdot 2^x + 81 = 2 \cdot 2^x + 17$$

$$\Rightarrow 2^{2x} - 20 \cdot 2^x + 64 = 0$$

$$\Rightarrow 2^{2x} - 16 \cdot 2^x - 4 \cdot 2^x + 64 = 0$$

$$\Rightarrow 2^x (2^x - 16) - 4 (2^x - 16) = 0$$

$$\Rightarrow (2^x - 4) (2^x - 16) = 0$$

The values of 2^x can't be 4 (log will be undefined), which implies The value of 2^x is 16.

Therefore, the common difference is $\log_4 (2^x - 9) - \log_4 2$

$$\Rightarrow \log_4 7 - \log_4 2 = \log_4 (\frac{7}{2})$$

The correct option is D

[VIDEO SOLUTION](#)

49.5

It is given that $5^{n-1} < 3^{n+1}$, where n is a natural number. By inspection, we can say that the inequality holds when n = 1, 2, 3, 4, and 5.

Now, we need to find the least integer value of m that satisfies $3^{n+1} < 2^{n+m}$

For, n = 1, the least integer value of m is 3.

For, n = 2, the least integer value of m is 3

For, n = 3, the least integer value of m is 4.

For, n = 4, the least integer value of m is 4.

For, n = 5, the least integer value of m is 5.

Hence, the least integer value of m such that for all the values of n, the equation holds is 5.3

[VIDEO SOLUTION](#)

50.468

We know that the number of factors of these two numbers is 15. We know that the factors of 15 are 1, 3, 5, and 15.

The number of factors of N is $(p+1) \cdot (q+1)$ (Where, $N = a^p \cdot b^q$, and a, b are prime numbers).

Hence, the value of N will be least when (p+1) and (q+1) are as close as possible and a, and b are the least distinct prime numbers.

Hence, $p+1 = 3 \Rightarrow p = 2$, and $q+1 = 5 \Rightarrow q = 4$, and the prime numbers a, and b are 2, and 3, respectively.

Hence, the lowest value of N is $N = 2^4 \times 3^2 = 144$, and the second lowest value of N is $N = 2^2 \times 3^4 = 324$.

Hence, the sum is $(144+324) = 468$

[VIDEO SOLUTION](#)

100+ CAT Quant Questions



51.9

It is given that $x^2 + bx + c = 0$ has two real roots. Let the roots of the equation be α, β . ($\alpha > \beta$)

Then, we can say that $\frac{1}{\alpha} - \frac{1}{\beta} = \frac{1}{3}$ Eq(1)

Similarly, $\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{5}{9}$ Eq (2)

Eq(2) can be written as $\left(\frac{1}{\alpha} - \frac{1}{\beta}\right)^2 + 2 \cdot \frac{1}{\alpha} \cdot \frac{1}{\beta} = \frac{5}{9}$

$$\Rightarrow \left(\frac{1}{3}\right)^2 + 2 \cdot \frac{1}{\alpha} \cdot \frac{1}{\beta} = \frac{5}{9}$$

$$\Rightarrow \frac{2}{\alpha \cdot \beta} = \frac{4}{9} \Rightarrow \frac{1}{\alpha \cdot \beta} = \frac{2}{9}$$

$$\Rightarrow \alpha \cdot \beta = \frac{9}{2}$$

We know that the product of the roots is equal to c, which implies $c = \frac{9}{2}$

We also know that the sum of the roots is equal to -b.

$$\Rightarrow \frac{1}{\alpha^2} + \frac{1}{\beta^2} = \left(\frac{1}{\alpha} + \frac{1}{\beta}\right)^2 - \frac{2}{\alpha \beta} = \frac{5}{9}$$

$$\Rightarrow \left(\frac{\alpha + \beta}{\alpha \beta}\right)^2 - \frac{4}{9} = \frac{5}{9}$$

$$\Rightarrow \left(\frac{\alpha + \beta}{\alpha \beta}\right)^2 = (1)^2$$

$$\Rightarrow \alpha + \beta = \pm \alpha \beta$$

Hence, the maximum value of b is $\frac{9}{2}$.

Hence, the maximum value of (b+c) is 9

[VIDEO SOLUTION](#)

52. C

It is given that a merchant purchases a cloth at a rate of Rs.100 per meter and receives 5 cm length of cloth free for every 100 cm length of cloth purchased by him.

Hence, the cost price of 105 cm clothes is 100 rupees.

It is also known that he marked the price of 100 cm clothes as 110 rupees, and gave a 5% discount, and he cheated his customers by giving 95 cm length of cloth for every 100 cm length of cloth purchased by the customers.

Hence, the selling price of 95 cm clothes is $110 \times \frac{19}{20}$ rupees.

Therefore, the selling price of 105 cm clothes is 115.5 rupees.

Hence, the profit is 15.5%

The correct option is C

[VIDEO SOLUTION](#)

53. D

Let the work done by Rahul, Rakshita, and Gurmeet be a, b, and c units per day, respectively, and the total units of work are W.

Hence, we can say that $7(a+b+c) < W$ (Rahul, Rakshita, and Gurmeet, working together, would have taken more than 7 days to finish a job).

Similarly, we can say that $15(a+c) > W$ (Rahul and Gurmeet, working together would have taken less than 15 days to finish the job)

Now, comparing these two inequalities, we get: $7(a+b+c) < W < 15(a+c)$

It is also known that they all worked together for 6 days, followed by Rakshita, who worked alone for 3 more days to finish the job. Therefore, the total units of work done is: $W = 6(a+b+c) + 3b$

Hence, we can say that $7(a+b+c) < 6(a+b+c) + 3b < 15(a+c)$

Therefore, $(a+b+c) < 3b \Rightarrow a+c < 2b$, and $9b < 9(a+c) \Rightarrow b < a+c$

$\Rightarrow a+b+c < 3b \Rightarrow 7(a+b+c) < 21b$, and $15b < 15(a+c)$

Hence, The number of days required for b must be in between 15 and 21 (both exclusive).

Hence, the correct option is D

[VIDEO SOLUTION](#)

54. C

It is given that the population of the town in 2020 was 100000. The population decreased by y% from the year 2020 to 2021 and increased by x% from the year 2021 to 2022, where x and y are two natural numbers.

Hence, the population in 2021 is $100000 \left(\frac{100-y}{100} \right)$.

The population in 2022 is $100000 \left(\frac{100-y}{100} \right) \left(\frac{100+x}{100} \right)$

It is also given that the population in 2022 was greater than the population in 2020 and the difference between x and y is 10.

Hence,

$$100000 \left(\frac{100-y}{100} \right) \left(\frac{100+x}{100} \right) > 100000, \text{ and } (x-y) = 10$$

$$\Rightarrow 100000 \left(\frac{100-y}{100} \right) \left(\frac{110+y}{100} \right) > 100000$$

$$\Rightarrow \frac{100-y}{100} \left(\frac{110+y}{100} \right) > 1$$

To get the minimum possible value of 2021, we need to increase the value of y as much as possible.

$$\text{Hence, } (100-y) \{ (100+y) + 10 \} > 10000$$

$$\Rightarrow 10000 - y^2 + 1000 - 10y > 10000$$

$$\Rightarrow y^2 + 10y < 1000$$

$$\Rightarrow y^2 + 10y + 25 < 1025$$

$$\Rightarrow (y+5)^2 = 1024 < 1025$$

$$\Rightarrow (y+5)^2 = 32^2$$

$$\Rightarrow y = 27$$

Hence, the population in 2021 is $100000 \times (100-27) = 73000$

The correct option is C

[VIDEO SOLUTION](#)

55. A

Let the volume of mixture A be 200 ml, which implies the quantity of cocoa in the mixture is 120 ml, and the quantity of sugar in the mixture 80 ml.

Similarly, let the volume of the mixture be 300 ml, which implies the quantity of coffee, and sugar in the mixture is 210, and 90 ml, respectively.

Now we combine mixture A, and B in the ratio of 2:3 (if 200 ml mixture A, then 300 ml of mixture B).

Hence, the volume of the mixture C is $(200+300) = 500$ ml, and the quantity of the sugar is $(90+80) = 170$ ml.

Now he mixes C with an equal amount of milk to make a drink, which implies the quantity of the final mixture is $(500+500) = 1000$ ml.

The quantity of sugar in the final mixture is 170 ml.

Hence, the percentage is 17%

The correct option is A

[VIDEO SOLUTION](#)

100+ CAT DILR Questions



56. C

Let us assume that A, B, C, D, and E weights are a, b, c, d, and e.

1st condition

$$\frac{(a+b+c)}{3} - \frac{(a+b+c+d)}{4} = x$$

2nd condition

$$\frac{(a+b+c+e)}{4} - \frac{(a+b+c)}{3} = 2x$$

Adding both the equations, we get:

$$\frac{(e-d)}{4} = 3x$$

$$\Rightarrow \frac{(e-d)}{4} = 3x \Rightarrow e-d = 12x$$

Given that $12x = 12 \Rightarrow x = 1$.

[VIDEO SOLUTION](#)

57. D

Let us assume the speed of the 1st boat is b, the 2nd boat is s, and the river's speed is r.

Let 'd' be the distance between A and B.

$$\Rightarrow d = 2(b+r) \text{ and } d = 3(b-r)$$

$$\Rightarrow b+r = d/2 \text{ and } b-r = d/3 \Rightarrow r = d/12 \text{ (subtracting both equations).}$$

Now, it is given that

$$\frac{d}{s+r} + \frac{d}{s-r} = 6$$

$$\Rightarrow \frac{d}{s + \frac{d}{12}} + \frac{d}{s - \frac{d}{12}} = 6$$

$$\Rightarrow 2ds = 6 \left(s^2 - \frac{d^2}{144} \right)$$

$$\Rightarrow 144s^2 - 48ds - d^2 = 0$$

Solving the quadratic equation, we get:

$$s = d \left(\frac{(48 + \sqrt{48^2 + 4(144)})}{2 \times 144} \right)$$

$$s = d \left(\frac{1}{6} + \frac{\sqrt{5}}{12} \right)$$

$$\Rightarrow \text{Required value of } \frac{d}{s+r}$$

$$= \frac{d}{\frac{d}{6} + \frac{\sqrt{5}d}{12} + \frac{d}{12}}$$

$$= \frac{12}{3 + \sqrt{5}} = \frac{(12)(3 - \sqrt{5})}{4}$$

$$= 3(3 - \sqrt{5})$$

[VIDEO SOLUTION](#)

58. 42

Given that the number of coins collected per week by two coin-collectors, A and B, are in the ratio 3: 4

Let us assume A collects 3c coins per week and B collects 4c coins per week.

Total number of coins collected by A in 5 weeks = $5 \times 3c = 15c$, which should be multiple of 7 $\Rightarrow c$ should be multiple of 7.

Total number of coins collected by B in 3 weeks = $3 \times 4c = 12c$, which should be a multiple of 24 $\Rightarrow c$ should be a multiple of 2.

So, the least possible value of c is $\text{lcm}(2,7) = 14$.

Coins sold by A in a week = $3c = 3 \times 14 = 42$.

VIDEO SOLUTION

59.340

Let us assume the initial stock of all the fruits is S.

Let us take we have 'b' and 'a' mangoes initially.

Stock of Mangoes = 40% of S = $2S/5$

The total number of fruits sold are Mangoes Sold + Apples Sold + Bananas Sold

$$= 2S/10 + 96 + 4a/10 = S/2 \text{ (Given)}$$

$$\Rightarrow S/5 + 96 + 2a/5 = S/2$$

$$\Rightarrow S = \frac{(4a + 960)}{3}$$

$$\Rightarrow \frac{4a}{3} + 320$$

'a' has to be a multiple of 3 for the above term to be an integer.

But 'a' has to be a multiple of 5 for $4a/10$ to be an integer.

$\Rightarrow$ The smallest value of 'a' satisfying both conditions is 15.

$$\Rightarrow \frac{4a}{3} + 320 = \frac{4(15)}{3} + 320 = 340$$

VIDEO SOLUTION

60.36

Let 'g' and 's' be the efficiencies of Gautam and Suhani. Let W is the total amount of work.

$$\Rightarrow g + s = W/20 \text{ (1 day work)} \text{ ---(1)}$$

Also Gautam doing only 60% $\Rightarrow 3g/5$ and Suhani doing 150% $\Rightarrow 3s/2$

$$\Rightarrow 3g/5 + 3s/2 = W/20 \text{ (1 day work)}$$

$$\Rightarrow g + s = \frac{3g}{5} + \frac{3s}{2}$$

$$\Rightarrow \frac{s}{g} = \frac{4}{5} \Rightarrow \text{Gautam is the more efficient person.}$$

Now, from the 1st equation

$$\Rightarrow g + \frac{4g}{5} = \frac{W}{20}$$

$$\Rightarrow \frac{9}{5}g = \frac{W}{20}$$

$$\Rightarrow g = \frac{W}{36}$$

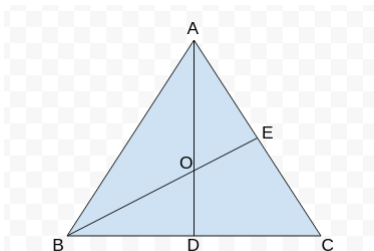
$\Rightarrow$ Gautam takes 36 days to finish the complete work.

VIDEO SOLUTION

100+ CAT VARC Questions



61.C



Given that $AB = AC \Rightarrow \text{Angle } C = \text{Angle } B \text{ (1)}$

AD and BE are altitudes $\Rightarrow$ they make 90 degrees with the sides

Angle AOB = 105 $\Rightarrow$ Angle EOD = 105 (Vertically Opposite Angles)

In quadrilateral DOEC

Angle C = $360 - 105 - 90 - 90 = 75 \Rightarrow \text{Angle } B = 75 \text{ (from 1)}$

We know that from the area of the triangle $AD * BC = BE * AC$

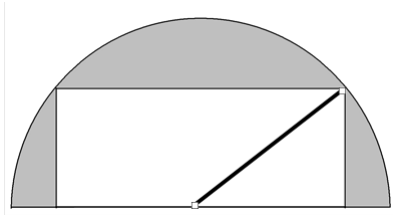
$$\Rightarrow \frac{AD}{BE} = \frac{AC}{BC} = \frac{2R \sin B}{2R \sin A} = \frac{\sin(75)}{\sin(30)} = 2 \sin(75) = 2 \cos(15)$$

$[\sin(x) = \cos(90-x)]$

VIDEO SOLUTION

62. A

Let us assume the length of the rectangle is 'l' and breadth of the rectangle is 'b'.



The radius, $l/2$ and b in the above diagram form a right-angled triangle.

$$\Rightarrow \left(\frac{l}{2}\right)^2 + b^2 = 2^2$$

We know that the area of the rectangle is $l \cdot b$, which can be obtained by considering 2 times the geometric mean of $\left(\frac{l}{2}\right)^2$ and b^2 .

Therefore, for the maximum area, the equality condition of AM-GM inequality should be satisfied

$$\Rightarrow \left(\frac{l}{2}\right)^2 = b^2 \Rightarrow l = 2b.$$

$$\Rightarrow l/b = 2/1.$$

VIDEO SOLUTION

63. 54

The sum of the interior angles of a polygon of 'n' sides is given by $(2n - 4) \times 90$, and the sum of the exterior angles of a polygon is 360 degrees.

So, the difference between them will be $120 \cdot n$

$$\Rightarrow (2n - 4) 90 - 360 = 120n$$

$$\Rightarrow 60n = 720 \Rightarrow n = 12.$$

We know that the number of diagonals of a regular polygon is $nC2 - n = 12C2 - 12 = 66 - 12 = 54$.

VIDEO SOLUTION

64. D

The given sequence can be written as:

$$1 \left(1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \dots\right) + \frac{1}{3} \left(\frac{1}{4} + \frac{1}{16} + \dots\right) + \frac{1}{9} \left(\frac{1}{16} + \frac{1}{64} + \dots\right) + \dots$$

We know that the sum of an infinite G.P. is $\frac{a}{1-r}$, where a is the first term and r is the common ratio.

$$\Rightarrow \text{The first term} = \frac{1}{1 - \frac{1}{4}} = \frac{4}{3}$$

$$\Rightarrow \text{The second term} = \frac{1}{3} \left(\frac{\left(\frac{1}{4}\right)}{1 - \left(\frac{1}{4}\right)}\right) = \frac{1}{9}$$

$$\Rightarrow \text{The third term} = \frac{1}{9} \left(\frac{\left(\frac{1}{16}\right)}{1 - \left(\frac{1}{4}\right)}\right) = \frac{1}{108}$$

Observing these three terms, we see that they are in G.P. with a common ratio of $\frac{1}{12}$

$$\Rightarrow \text{Sum of this infinite G.P.} = \frac{\left(\frac{4}{3}\right)}{1 - \left(\frac{1}{12}\right)} = \frac{16}{11}$$

VIDEO SOLUTION

65. A

The first series goes as follows:

46, 54, 62, 70, 78, 86, 94, 102,

The second series goes as follows:

98, 102, 106, 110, ...

The first common term is 102 (first term of the common terms) and the common difference between them will be $\text{hcf}(4, 8) = 8$

$\Rightarrow$ The required sequence is 102, 110, 118, (last term should be less than 468 (100th term of second series))

$$\Rightarrow 102 + (n-1)(8) \leq 498$$

$$\Rightarrow n \text{ is less than or equal to } 50.5 \Rightarrow n = 50$$

Using the summation of A.P. formula:

$$\text{Required sum} = \frac{n}{2} (2a + (n-1)d) = \frac{50}{2} (2 \times 102 + 49 \times 8) = 14900$$

VIDEO SOLUTION

CAT Expected Questions PDF



66.3

Given that $f(3x + 2y, 2x - 5y) = 19x$.

Let us assume the function $f(a,b)$ is a linear combination of a and b .

$$\Rightarrow f(3x+2y, 2x-5y) = m(3x+2y) + n(2x-5y) = 19x$$

$$\Rightarrow 3m + 2n = 19 \text{ and } 2m - 5n = 0$$

Solving we get $m = 5$ and $n = 2$

$$\Rightarrow f(a,b) = 5a+2b$$

$$\Rightarrow f(x,2x) = 5x + 2(2x) = 9x = 27 \Rightarrow x = 3.$$

VIDEO SOLUTION